

MICROGRID ISLANDING AND SCENARIO ANALYSIS

MATLAB/Simulink Simulation Report

Candidate Name: Spandan Mondal
Email: spandanmondal66@gmail.com
[LinkedIn](#)
[GitHub](#)

1. Objective

To develop and simulate a PV-BESS based microgrid in MATLAB/Simulink and study its response during grid-connected and islanded operation. The simulation is used to examine the effects of grid disconnection, reduction in PV generation, BESS power support and limitation, frequency variation, SOC change, power imbalance, and load shedding.

2. Introduction

A microgrid is a localized electrical system containing generation sources, energy storage systems, and electrical loads that can operate either in connection with the utility grid or independently in islanded mode.

The model developed in MATLAB/Simulink consists of a photovoltaic (PV) source, battery energy storage system (BESS), utility grid, and electrical load. The model is used to observe the system response before and after islanding.

Following islanding, the PV output is reduced significantly, creating a generation deficit. The BESS is used to provide power support, subject to its maximum power limit. A load-shedding mechanism is also implemented to reduce the electrical load when specified operating conditions are detected.

The frequency response, power balance, BESS power, SOC, PV power, grid power, and load power are monitored using Simulink Scope blocks.

3. System Configuration

The simulated microgrid consists of the following major components:

- Photovoltaic (PV) generation
- Battery Energy Storage System (BESS)
- Utility grid
- Electrical load
- Frequency-dynamics model
- SOC calculation subsystem
- Load-shedding logic
- Power-balance calculation
- Simulink Scope blocks for monitoring system variables

The basic operating sequence is:

Grid-connected operation → Islanding at 2 s → PV power disturbance → BESS support → Load shedding → Islanded operation

4. Simulation Parameters

Parameter	Value
PV rated power	100 kW
Initial load	100 kW
Grid power before islanding	50 kW
Grid power after islanding	0 kW
Islanding time	2 s
PV power after disturbance	10 kW
Load after load shedding	90 kW
Initial BESS SOC	60%
BESS discharge saturation	80 kW
Nominal frequency, f_0	50 Hz
Inertia constant, H	2.5
Damping coefficient, D	1
Frequency load-shedding threshold $f < 49\text{Hz}$	

5. Scenario Definition

5.1 Scenario 1 – Grid-Connected Operation

Initially, the microgrid operates in grid-connected mode. The utility grid supplies 50 kW, while the PV source operates at 100 kW and the load is 100 kW.

This condition is used as the initial operating point for the simulation before the grid is disconnected.

Observation:

The microgrid operates in grid-connected mode before 2 s. During this period, the utility grid supplies 50 kW while the PV source supplies 100 kW to the system. The load demand is maintained at 100 kW. The system remains connected to the utility grid and operates under the initial operating condition.

5.2 Scenario 2 – Islanded Operation with PV Disturbance

At $t = 2\text{s}$, the utility grid is disconnected from the microgrid. Consequently, the grid contribution changes from 50 kW to 0 kW.

At the same time, the PV generation is reduced from 100 kW to 10 kW. The BESS responds to the deficit by increasing its discharge power, up to the 80 kW limit used in the model.

The load-shedding logic is used to reduce the load from 100 kW to 90 kW when the specified control conditions are satisfied.

Observation:

At 2 s, the utility grid is disconnected and the microgrid enters islanded operation. The grid power reduces to 0 kW, while the PV power decreases to 10 kW. The BESS responds to the resulting power deficit and reaches its specified saturation limit of 80 kW.

6. Grid-to-Island Transition

The grid disconnection is modeled at $t = 2$ s. Before this instant, the microgrid receives power from the utility grid. After $t = 2$ s, the grid power becomes zero and the microgrid must operate independently.

The grid disconnection at 2 s is used to observe how the microgrid responds when it has to operate without grid support.

The transition can be represented as:

$$P_{grid} = \begin{cases} 50 \text{ kW}, & t < 2 \text{ s} \\ 0 \text{ kW}, & t \geq 2 \text{ s} \end{cases}$$

The Scope output for grid power is used to verify this transition.

7. PV Power Disturbance

The PV generation is initially maintained at 100 kW. After islanding, the PV output is reduced to 10 kW.

Thus,

$$P_{PV} = \begin{cases} 100 \text{ kW}, & t < 2 \text{ s} \\ 10 \text{ kW}, & t \geq 2 \text{ s} \end{cases}$$

The reduction in PV output leaves less generation available to supply the load. The BESS therefore has to provide additional power, and the load-shedding control can reduce the demand when its conditions are met.

8. BESS Model and Power Limitation

The BESS is included to provide power support during islanded operation. Its power output is limited using a Saturation block to represent the maximum available discharge power.

The saturation limit used in the simulation is:

$$|P_{BESS}| \leq 80 \text{ kW}$$

Under the adopted sign convention, the simulated BESS power reaches:

$$P_{BESS} = -80 \text{ kW}$$

when the discharge limit becomes active.

The Saturation block limits the BESS output to 80 kW. Once this value is reached, any remaining deficit cannot be supplied by increasing the BESS power further.

9. BESS State of Charge Calculation

The initial BESS state of charge is set to:

$$SOC_0 = 0.6$$

or

$$SOC_0 = 60\%$$

The SOC is calculated by integrating the battery power with the appropriate scaling factors.

The implemented processing chain is:

BESS Power → **Gain** → **Gain** → **Integrator** → **SOC Scaling** → **Scope**

The Integrator initial condition is set to 0.6.

The SOC output is monitored throughout the simulation to observe the effect of BESS operation on stored energy.

10. Power-Balance Calculation

The power-balance signal is used to determine the difference between the available generation and the load demand.

The implemented expression is:

$$P_{balance} = P_{PV} + P_{grid} - P_{BESS} - P_{load}$$

The sign of P_{BESS} is interpreted according to the sign convention used in the Simulink model.

For example, after islanding, with:

$$\begin{aligned} P_{PV} &= 10 \text{ kW} \\ P_{grid} &= 0 \text{ kW} \\ P_{BESS} &= -80 \text{ kW} \end{aligned}$$

and

$$P_{load} = 100 \text{ kW}$$

the resulting balance is:

$$\begin{aligned} P_{balance} &= 10 + 0 - (-80) - 100 \\ P_{balance} &= -10 \text{ kW} \end{aligned}$$

After load shedding reduces the load to 90 kW, the corresponding balance becomes:

$$\begin{aligned} P_{balance} &= 10 + 0 - (-80) - 90 \\ P_{balance} &= 0 \text{ kW} \end{aligned}$$

The remaining mismatch shows why reducing the load is useful when the available PV and maximum BESS output are not sufficient to meet the full demand.

11. Frequency-Dynamics Model

The frequency behavior of the islanded microgrid is represented using a simplified swing-equation model:

$$2H \frac{df}{dt} = P_{balance} - D(f - f_0)$$

where:

- H = inertia constant
- D = damping coefficient
- f = system frequency
- f_0 = nominal frequency
- $P_{balance}$ = power imbalance

The values used are:

$$\begin{aligned} H &= 2.5 \\ D &= 1 \\ f_0 &= 50 \text{ Hz} \end{aligned}$$

Therefore,

$$\frac{df}{dt} = \frac{1}{2H} [P_{balance} - D(f - f_0)]$$

or

$$\frac{df}{dt} = 0.2[P_{balance} - (f - 50)]$$

The output of the Integrator provides the system frequency, with an initial condition of 50 Hz.

12. Load-Shedding Logic

A load-shedding control is added to change the load from 100 kW to 90 kW when either of the defined operating conditions is detected.

Two conditions are considered:

Frequency condition

$$f < 49 \text{ Hz}$$

BESS/SOC condition

The BESS/SOC operating condition is monitored using a relational operator.

The outputs of these conditions are connected to an OR gate.

$$Condition_{LS} = Condition_f \text{ OR } Condition_{SOC}$$

The OR-gate output controls a Switch block.

The Switch selects between:

- **100 kW:** normal load
- **90 kW:** shed-load condition

The resulting load is therefore:

$$P_{load} = \begin{cases} 100 \text{ kW,} & \text{normal condition} \\ 90 \text{ kW,} & \text{load-shedding condition} \end{cases}$$

The final Switch output is connected to the load input.

13. Simulation Results

Parameter	0s	2s	5s
Grid power	50 kW	-	0 kW
PV power	100 kW	-	10 kW
BESS power	-50	-	-80
Load power	100 kW	100 kW	100 kW
Frequency	50 Hz	82.968 Hz	63.5813 Hz
BESS SOC	60%	60.0139%	60.0472%
Power balance	100 kW	-	-10 kW

14. Scenario Analysis

14.1 Scenario 1 – Grid-Connected Operation

From 0 to 2 s, the microgrid remains connected to the utility grid. The PV source is set to 100 kW and the load is 100 kW, while the grid supplies 50 kW. This operating period is used as the initial condition before the islanding event.

14.2 Scenario 2 – Islanded Operation

At $t = 2$ s, the grid is disconnected and its power contribution becomes 0 kW. The PV output is also reduced from 100 kW to 10 kW. This causes a large change in the power balance, and the BESS responds by supplying power to the islanded system.

The BESS output reaches the 80 kW saturation limit. The load-shedding logic is then used to reduce the load from 100 kW to 90 kW when the defined control condition is satisfied.

14.3 Effect of PV Power Reduction

The reduction of PV power from 100 kW to 10 kW is the main generation-side disturbance after islanding. With less PV power available, the BESS has to supply a larger portion of the load. Since the BESS is limited to 80 kW, the remaining mismatch has to be addressed by reducing the load.

14.4 Effect of BESS Saturation

The BESS reaches its specified discharge limit of 80 kW during islanded operation. This shows that the battery is being used at its maximum allowed power in the simulated condition. Increasing the battery output beyond this value is prevented by the Saturation block.

14.5 Effect of Load Shedding

The load-shedding control changes the load from 100 kW to 90 kW when either of the defined conditions becomes active. The frequency condition and the BESS/SOC condition are combined using an OR gate, and its output controls the Switch block. This allows the load to be reduced when the islanded system requires additional support.

14.6 Frequency Response

The frequency remains at its initial value of 50 Hz during the initial operating period and changes after the system enters islanded operation. The change is produced by the power imbalance in the simplified frequency-dynamics model. The BESS response and load-shedding action also affect the resulting frequency.

14.7 SOC Response

The BESS SOC starts at 60% and changes as the battery supplies power to the microgrid. The change in SOC is obtained from the battery power through the scaling and integration blocks implemented in the model. The SOC output is used to monitor the battery condition during the simulation.

15. Discussion

The simulation shows the response of the microgrid when the utility supply is removed and the system continues in islanded operation. Before 2 s, the grid supplies 50 kW while the PV source supplies 100 kW to the 100 kW load. At 2 s, the grid contribution becomes zero and the PV output is reduced to 10 kW.

The loss of grid power together with the reduction in PV generation produces a significant change in the power balance. The BESS responds by increasing its discharge power and reaches the specified limit of 80 kW. Since the battery cannot provide more than this value, the remaining mismatch is handled by the load-shedding control.

The load-shedding logic reduces the load from 100 kW to 90 kW when the defined frequency or BESS/SOC condition is satisfied. This reduces the demand on the islanded system and helps to reduce the remaining power mismatch.

The frequency response reflects the change in the power balance after islanding. Similarly, the SOC changes from its initial 60% value as the BESS supplies power. These signals provide a way to observe the response of the microgrid and the effect of the control actions during islanded operation.

Overall, the simulation shows that the ability of the microgrid to continue operating after islanding depends on the available PV generation, the power that can be supplied by the BESS, and the amount of load that can be disconnected when required.

16. Key Results

The main observations from the simulation are:

1. The utility grid is disconnected at $t = 2$ s, causing the grid power to change from 50 kW to 0 kW.

2. The PV output is reduced from 100 kW to 10 kW after the disturbance.
 3. The BESS responds to the resulting power deficit and reaches its 80 kW discharge limit.
 4. The load-shedding control changes the load from 100 kW to 90 kW when the specified control condition is activated.
 5. The frequency changes from its initial 50 Hz value as a result of the power imbalance during islanded operation.
 6. The BESS SOC is initially set to 60% and changes according to the battery power during the simulation.
 7. The power-balance signal indicates the mismatch between the available generation, BESS contribution, and load demand.
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17. Conclusion

A PV-BESS based microgrid was modeled and simulated in MATLAB/Simulink for both grid-connected and islanded operation. The model was used to study the effect of grid disconnection, PV power reduction, BESS power limitation, frequency variation, SOC change, and load shedding.

At $t = 2$ s, the utility grid was disconnected and the PV output was reduced from 100 kW to 10 kW. The BESS responded to the resulting power deficit and reached its 80 kW power limit. The load-shedding logic was then used to reduce the load from 100 kW to 90 kW under the defined control conditions.

The simulation therefore provides a basic demonstration of how BESS support and load shedding can be used to respond to disturbances during islanded operation. The frequency, SOC, power-balance, PV, grid, BESS, and load signals were used to evaluate the response of the modeled microgrid.

18. Software Used

- MATLAB
 - Simulink
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19. References

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